ANNAMACHARYA UNIVERSITY

New Boyanapally, Rajampet, Annamayya District, A.P. 516126.

ANNAMACHARYA INSTITUTE OF TECHNOLOGY AND SCIENCES

Department : Computer Science & Engineering

ASSIGNMENT SHEET

Name of the student: Ch. Vena prasad
Year & Sem: III – I
Subject: Computer Networks

HT. No: 24ARCSE400
Section: G
Marks:

(Start writing from here)

(1.) Explain OSI network architecture and explain the functionalities of every layer in detail.

A: **OSI network architecture:**

The OSI model (minus the physical medium) is shown in the below figure. The model is based on a proposal developed by the International Standards Organization (ISO) as a first step toward international standardization of the protocols used. It is called the ISO OSI (open System Interconnection) Reference model because it deals with Connecting open Systems - that is, systems that are open for Communication with other Systems. We will Call it the OSI model for short.

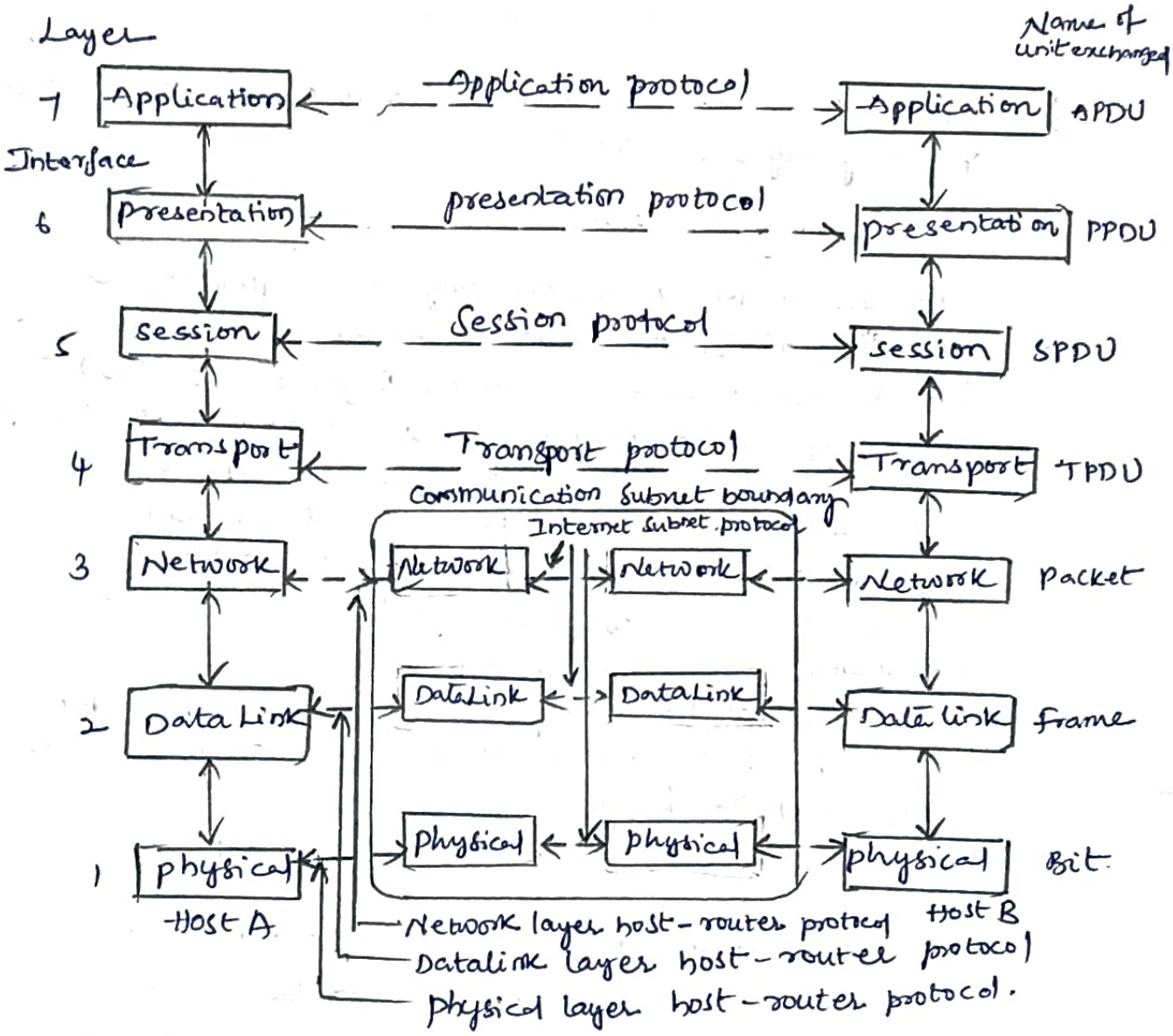
The OSI model has ~~been~~ seven layers. The principles that were applied to arrive at the seven layers can be briefly Summarized as follows:

1. A layer should be created where a different abstraction is needed.
2. Each layer should perform a well-defined function.
3. The function of each layer should be choosen with an eye toward defining internationally standardized protocols.
4. The layer boundaries should be chosen to minimize the information flow across the interfaces.
5. The no. of layers should be large enough that the distinct functions need to be thrown together.

Three concepts are central to the OSI model:

1. Services
2. Interfaces
3. Protocols.

Probably, the biggest contribution of the OSI model is that it makes the distinction between these three concepts explicits. Each layer performs some services for the layer above it.



The Data Link Layer:

The main task of the datalink layer is to transform a raw transmission facility into a line that appears free of undetected transmission errors. It does so by masking the real errors so the network layer doesnot see them.

Broadcast networks have an additional issue in the data link layer: how to control access to the shared channel. A special sublayer of the data link layer, the medium access control sublayer, deals with this problem.

The Network Layer:

The network layer controls the subnet. A key design issue is determining how packets are routed from source to destination. Routes can be based on static tables that are "wired into" the network and rarely changed. More generally, the quality of service provided (delay, transmit time, jitter, etc.) is also a network layer issue.

The Transport Layer:

The transport layer provides end-to-end communication b/w applications running on different devices. It responsible for making sure data reaches the correct application and, depending on the protocol reliably.

The Session Layer:

The session layer establishes, manages and terminates communication sessions b/w applications. A session is a logical connection b/w two applications.

Functionalities:

* Establishes communication sessions
* Maintains session
* provides dialog control b/w devices.

The presentation Layer:

The presentation layer is responsible for the format and representation of data. It makes sure that data sent by one system can be understand by another system.

Functionalities:

* performs data translation b/w different formats.
* performs encryption and decryption.
* Handles data representation and formatting.

The application Layer :

The application layer is the top layer of the OSI model. It provides network services directly to applications used by the ~~user~~ end user.

Functionalities:

* Performs data or provides network services directly to user applicant.
* Supports web browsing
* Supports file transfer.
* Supports email services.

The physical Layer:

The physical layer is the lowest layer of the OSI model,

Functionalities:

* It is responsible for transmitting raw bits (0s and 1s) from one device to another through a physical medium
* It defines the electrical, mechanical and physical char of the connection.
* It determines how bits are represented as electrical, optical, or radio signals.
* It defines the physical topology of the network.

Ex: Ethernet cables, Radio signals, etc.

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(2) a) Describe wide area networks and its components.

A: A WAN (Wide Area Network) spans large geographical area, often a country or continent. We will begin our discussion with the example of a company with branch offices in different cities. In most WANs, the subnet consists of two distinct components: Transimission lines and switching elements.

1. Transmission Lines:
They move bits between machines. They can be made of copper wire, optical fiber, or even radio links. Most companies do not have transmission lines lying about, so instead they lease the lines from a telecommunications company.

2. Switching Elements:
These are just specialized computers that connect two or more transmission lines. When data arrives on an incoming line, the switching element must choose an outgoing line on which to forward it. These switching computers have been called by various names in the past

b) Explain briefly about the original ARPANET design.

The original ARPANET design was developed in response to the need for a reliable communication network. During the Cold War, the U.S. Department of Defense wanted to command-and-control network that could service a nuclear war. The existing telephone network was considered vulnerable because destroying a few important switching offices could fragment the network.

Main features of the original ARPANET design.

* The subnet consisted of minicomputers called Imps.
* Imps were connected using 56-Kbps transmission lines.
* For high reliability, each Imp was connected to at least two other IMPs.
* Every packet contained lines, and if IMPs were destroyed, packets could be automatically rerouted through alternative paths.
* Each node consisted of an IMP and a host connected by a short wire.
* A host could send messages of up to 8063 bits to its IMP.
* The IMP divided these messages into packets of at most 1008 bits.
* Packets were forwarded independently toward the destination.
* Each packet was recieved completely before being forwarded, making ARPANET the first electronics, store-and-forward packet-switching network.

Thus, the original ARPANET design emphasized packet switching, reliability, alternative paths, and store-and-forward transmission.

(3.) What is the need of framing? Explain character stuffing and bit stuffing for framing.

Ans: Need of framing:

The physical layer provides a new bit stream to the data link layer, The data link layer needs to divide this continuous bit stream into discrete frames so that data can be handled properly. The data link layer breaks the bit stream into frames, calculates a checksum for each frame, and includes the checksum with the frame. At the destination, the checksum is recalculated to detect transmission errors-

Character stuffing:

It is a framing technique in which, phyto special characters are used to indicate the begining and end of a frame.

* A special byte called a FLAG is used as a delimiter.

Basic structure:

FLAG | DATA | DATA | DATA | FLAG

* The recieved may incorrectly think that the middle FLAG represents the end of the frame.

* The recieved knows that
  FLAG First → start of frame
  Last FLAG → end of frame

Ex: What if the FLAG character appears naturally inside the data?

FLAG | A | B | FLAG | C | D | FLAG

Ex: 1. Using ESC character

To solve this problem, the sender inserts an ESC character before an accidental FLAG that appears in the data.

Before stuffing:

FLAG | A | B | FLAG | C | D | FLAG

After stuffing:

FLAG | A | B | ESC FLAG | C | D | FLAG

* The recieved recognizes ESC + FLAG as data rather than as a frame boundary, and removes the ESC.

Bit stuffing:

It is similar to character stuffing, but instead of inserting an extra character / byte, the sender inserts an extra bit.

It is commonly used with a special flag bit patter:
01111110

Ex: Original data

01111110

* It is occured as data, it would look exactly like the FLAG.
* The sender stuffs a '0' after every five consecutive 1s : 011111010
* The reciever detects five consecutive 1 as followed by 0 and removes that stuffed 0.

So,

Sender

01111101 0
↑
stuffed 0

Reciever

01111110

(4) Explain guided transmission media: twisted pair and coaxial cable.

Ans: Guided Transmission media:

The purpose of the physical layer is to transport bits from one machine to another. Various physical media can be used for the actual transmission. Each one has its own niche in terms of band width, delay, cost and ease of installation and maintainance. Media are roughly grouped into guided media, such as copper wire and fiber optics, and unguided media, such as terrestrial wireless, satellite, and lasers through the air.

#Twisted pair:

It is one of the oldest and most commonly used transmission media. It consists of two insulated copper wires twisted together like a DNA molecule.

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* The wires are twisted to reduce interference and noise. Data is usually transmitted as the difference in voltage b/w the two wires, which provides better protection against external noise.
* Twisted pair can carry both analog and digital data. Its band width depends on the wire thickness and transmitted distance. It is widely used because it provides good performance at low cost.
* Twisted pair cables are three types.
  (i) category 5 (cat 5)
  (ii) Category 6 (cat 6)
  (iii) Category 7 (cat 7)

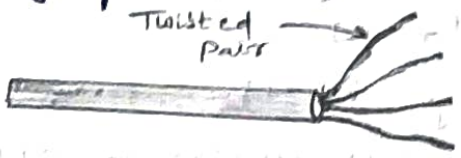


Fig: Category 5 UTP cable with four twisted pairs.

Coaxial Cable:

A coaxial cable (coax) is a common transmission medium. It has better shielding and higher band width than twisted-~~cable~~ pair cable, so it can transmit data over longer distances at higher speeds.

* A coaxial cable consists of:
  → A copper wire at the centre.
  → An insulating layer around it.

→ A metallic braided shield around the insulator.
→ An outer protective plastic sheath.

They are 2 common types of coaxial cables:

(i) 50-ohm coaxial cable

(ii) 75-ohm coaxial cable.

It can support ~~bous~~ band widths of upto a few GHz. It was previously used for long distance telephone lines, but fiber optics have largely replaced it.

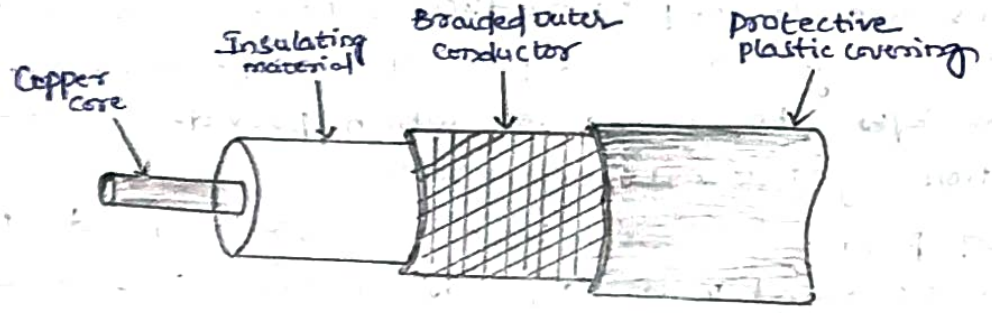


Fig: A coaxial cable.

(5.) Given 1101011011 data frame and generator polynomial $G(x) = x^4 + x + 1$, derive the transmitter frame.

Given Data frame: 1101011011

Generator $G(x) = x^4 + x + 1$

Generator in binary: 10011

Since the generator has degree 4, append 4 zeros:

11010110110000

Module-2 Division:

```
          10011
        ---------
10011 ) 11010 1101 1000
          10011
          -----
          10011
          10011
          -----
          00000
             ↓
          10010
          10011
          -----
          00101
             ↓
          10100
          10011
          -----
          00111
             ↓
           1110
```

So, the remainder = 1110

∴ Original Data = 11010 1101

Remainder (CRC) = 1110

Transmitted Frame = 11010 1101 1110 → 11010110111110